While most industry robots are carefully tuned and limited to work in highly structured environments in practice, I believe that more elegant solutions are needed to empower intelligent agents to work and interact with people. At UW RSE Lab, I have focused on robot manipulation as my research topic.

Perception allows robots to reason about the surrounding environment. Specifically, for object pose estimation, I have investigated training Deep Iterative Matching Network (DeepIM [1]) with synthetic data only. As a novel pose refinement method, DeepIM requires a significant number of real-world data with ground truth labels. While synthesizing images reduces the heavy work of labeling, training with rendered images meanwhile presents a challenge to generalization in the real world. So, I proposed to modify its training procedure by randomizing pose distributions and generating realistic occlusions in the training data. Moreover, I built a DeepIM tracking system that propagates the pose estimate from the previous frame to the current frame, which achieves state-of-the-art pose tracking accuracy in the YCB-Video dataset. Later experiments demonstrate that DeepIM is able to track robot hand and object poses under heavy occlusions.

Planning allows robots to find feasible trajectories to manipulate target objects. While grasp planning in task space and motion planning in configuration space are often tackled separately, I have delved into an integration of these two planning problems. After discussions with Dr. Yu Xiang at NVIDIA, we proposed a novel method [2] to integrate a grasping pipeline, namely grasp synthesis, grasp selection, and motion planning. Intuitively, humans do not have a determined and perfect grasp in mind before reaching, which allows re-planning in dynamic situations. Similarly, during motion planning, we apply online learning techniques to select goal configurations for grasping and introduce a grasp refinement procedure based on surface fitting. Overall, our method generates robust and efficient motion plans for grasping in cluttered scenes.

Control allows robots to operate safely and precisely in cluttered scenes. I have analyzed the performance of classical control, model-predictive control, reinforcement learning, and imitation learning in the Bullet simulation engine. This initial exposure taught me the limitation in generalization, sample efficiency, and hyperparameter sensitivity of naively applying a learning algorithm to control in partially observable environments.

At HRI, there are several research projects that interest me. I also believe HRI’s focus on scale and novelty of projects would help me derive principled robotic methods to assist people in need.

[1] https://liruiw.github.io/perception.html
[2] https://liruiw.github.io/planning.html